



Lab: Data Exploration using Python

In this Lab, you'll practice your data exploration skills as you progress through the Extract, Wrangle and Analyze phases of the Data Analytics Workflow. In this lab, you have 3 parts to complete. You'll start with **guided solo practice, followed by individual work on the same data set and digging deeper.**

You will have half a day of class to complete this Lab, and it **requires completion of all parts below.**

Coffee shop (time: half day)

The Scenario

In recent years, the appearance of new coffee shops has grown enormously.

Your client is the owner of a chain of coffee shops in New York City. They have provided a data set of around 50,000 transactions and would like a summary of their sales data.¹

There are

A. Guided solo practice

In this first section, follow along with the first part of the Jupyter notebook ('Coffee shop data') to clean and explore the data. Your focus will be on answering three questions:

1. Which products offer the best margins?
2. Are there any particular customer groups spending more?
3. Which products generate the most sales?

B. Individual data exploration

In this second section, explore the coffee shop data to answer these questions:

- What kinds of products generate the most profits?
- What kinds of customers generate the most profit ratio? Can you explain why?

¹ The data was created by IBM and is on [Kaggle](#)



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- Which segment of its customer base should the coffee shop target a marketing campaign towards?

C. Dig Deeper

Additional questions that you can also investigate:

- Are there better or worse days/times in terms of sales?
- Does customers' willingness to pay for merchandise vary by generation?
- Do customers who have been customers for longer tend to also spend more?

D. Incorporating PowerBI for your Visuals

Create **at least 5 different visualizations** (PowerBI) and data summaries to support your analysis.

You are required to select any 5 of the charts you created in jupyter notebook and replicated it in PowerBI.

- Import your csv to PowerBI and create further visualizations

Deliverables for the whole Lab

1. Complete the Jupyter Notebook (*.ipynb) with the required Python code filled in
2. A completed Word / Google Doc report of 1 page describing the results of your coffee shop analysis:
 - a. What is the problem you are trying to solve / question you are trying to answer?
 - b. What are the limitations of the data you have?
 - c. What key insights have you gained from the data?
 - d. So what? What actions can people take as a result of your analysis?

Process

Your finished deliverable should include evidence of:

- **Data Cleaning**
 - The report should mention any missing data you encountered and what you decided to do with it.
 - It should also mention any data processing you underwent, for example



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extracting text from within other text (finding coffee bean products).

- Exploring the Data
 - Which insights and visualizations were most informative, and how did they differ from what you expected?

Rubric

The purpose of this assignment is to assess your Python proficiency. Instructors will evaluate student skill based on the following rubric:

	Incomplete	Doesn't Meet Expectations	Meets Expectations
Jupyter Notebook and PowerBi workbook	Notebook not submitted	Notebook submitted but not completed	Notebook completed and submitted
Coffee shop part A <i>Data Cleaning</i>	Data not cleaned/modified in any way	Data is partially cleaned (e.g. missing values dropped without justification)	Data has been cleaned and decisions have been justified in the report
Coffee shop part B <i>Exploring the Data</i>	No exploration of the data has been performed	Some analysis has been done but is incomplete (e.g. only one question attempted)	Analysis has been performed and results communicated for all questions, with at least 4 supporting visual each.
Coffee shop Part C <i>Dig Deeper</i>	No analysis of the data has been performed	Some analysis has been done but is incomplete (e.g. only one question attempted)	Analysis has been performed and results communicated for all questions, with at least 1 supporting visual each.

You must receive a score of at least **Meets Expectations** in all categories (except the optional ones) to pass this Lab.

Good luck and have fun!